

NextGen Exoskeleton Navigator Technical Report
NextGen Exoskeleton Navigator Security Threat Exercise
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Executive Summary

The NextGen Navigator (NGEN) system is a state-of-the-art exoskeleton system with capabilities of manned, unmanned, and autonomous modes with the means to support security, research, and engineering personnel. As the pinnacle of NextGen technology, the importance of identifying and mitigating threats cannot be understated in their deployment groups.

Threat identification, modelling, and mitigation is an essential part of the secure software development lifecycle (SSDLC) and should be conducted continuously throughout the inception, production, and maintenance of the NextGen Exoskeleton Navigator system. Modern threats are dynamic, can be unique, and present uncomfortable situations for organizations. This can cause a sense of paranoia to be induced in personnel and users. Proper handling of cybersecurity awareness training, information assurance training programs, and adoption of zero-trust architecture throughout the organization should be of the highest priority as its success sets a fundamental foundation for future security focused mindsets. With this, each threat presented has been appended with the vulnerability in which the threat takes advantage of, and the mitigation techniques or security controls to be emplaced to deter their designated threats. It is essential all threats, even innocuously seeming ones are taken seriously and treated as potential vectors for asset damage and business costs.

In conclusion, the NextGen Exoskeleton Navigator System is the culmination of the greatest minds of our time to envision a product capable of propelling NextGen even further ahead of industry competition all while revolutionizing research, autonomous drone use, and security implementation. This report provides a detailed breakdown of fundamental or critical threats that the NextGen Navigator Exoskeleton and their deployments face while enabling stakeholders to understand the vulnerabilities and countermeasures used to mitigate them deployments face in a deployment environment.

Discussion

Threat and Vulnerability Review

Threats are any action that can cause damage or comprise an asset. Threats span a broad range of specific types from physical threats to malicious software threats, and as such can be complicated to clearly identify, rate, and properly mitigate. Threats can originate from an individual within an organization, an internal threat actor, or groups such as competitors and hacking organizations. Threats, or threat actors, take advantage of vulnerabilities to achieve their goal of exploiting a system and exfiltrating data or damaging assets. Vulnerabilities are flaws or a weakness inherent in a system or accidentally built into it by developers. When a threat exploits a vulnerability, a security breach occurs and the tenets of cybersecurity, confidentiality, availability, and integrity may be impacted. From these loose definitions, the presented threats, vulnerabilities, and their security controls have been created to facilitate the identification and mitigation of vulnerabilities.

Security Control Review

Security controls are tools, safeguards, or countermeasures used to protect assets. These include physical barriers, software barriers, prevention techniques, detection techniques, recovery plans, and many more. In essence, if it protects an asset and mitigates a vulnerability, it is a security control. Vulnerability mitigation and asset protection is a complicated task as every organization will have unique challenges, as well as common place ones. This dynamic threat environment presents a unique opportunity for security professionals to be creative and develop new forms of asset protection. From this perspective it is important to understand that threats cannot be mitigated, only the vulnerability they take advantage of can be, and the risk of its realization will always be non-zero.

Security Threats

Security Threat 1

Security Threat 01 was recognized as a threat to NextGen systems due to the scope of our operating theaters. Cyber-physical attacks are singular or tandem attacks that operate with the intention to destroy critical business functions and exfiltrate sensitive data. Vulnerabilities attached to this threat are foremost physical in nature, with physical security countermeasures being the most effective to deter such threat actors. With this, security controls such as armed personnel and physical separations between potential attackers and NextGen assets are the most efficient means of deterrence.

Sec. Threat 1

Threat ID	Threat(s)	Vulnerabilities	Security Controls
T01	Cyber-physical Attack – Physical points of entry to Drone Operations Center and Barrack’s under-protected and unmonitored. Threat actors could easily gain unauthorized access to Exoskeleton systems.	Inadequate physical security controls.	1. Implement physical access control countermeasures such as security personnel and physical barriers. 2. Install alarm systems and monitor points-of-entry to DoC, CC, and Barrack’s.

Security Threat 2

Security Threat 02 has been identified and recorded as a critical threat due to the nature of NGEN systems and their dynamic addressing schemes when deployed within field environments. The threat of internet-protocol spoofing takes advantage of poor internal security countermeasures and a lack of network-edge filtering devices. Security controls such as the encryption of data-in-transit

and security systems such as intrusion detection / prevention systems or firewalls are the most efficient means at mitigating the vulnerability utilized by Security Threat 02.

Sec. Threat 2

Threat ID	Threat(s)	Vulnerabilities	Security Controls
T02	IP Spoofing Attack – NGEN IP addresses are not static and are therefore open to being spoofed. Threat actors could easily pose as a NGEN Exoskeleton on the network and exfiltrate sensitive information.	Lack of network-edge filtering devices and data-in-transit encryption.	1. Introduce internal network filtering on critical convergence points between networks and sites. 2. Utilize an encryption standard to secure internal network communications.

Security Threat 3

Security Threat 03 is the use of a brute force attack on NGEN systems and login interfaces by means of automated cracking software. This attack takes advantage of a lack of user access control lockouts and multi-factor authentication techniques. Given the linearity of this threat and the vulnerabilities it takes advantage of, introducing multi-factor authentication and a maximum login attempt policy for user accounts is the most efficient means to mitigate this threat.

Sec. Threat 3

Threat ID	Threat(s)	Vulnerabilities	Security Controls
T03	Password Brute Force Attack – NGEN systems do not lock after a designated number of login attempts. Threat actors could conduct a brute force attack against exoskeleton system passwords to gain access.	No user account control password lockout policy implemented.	1. Introduce multi-factor authentication (MFA) to NGEN systems. 2. Enforce a maximum login attempt policy on NextGen systems.

Security Threat 4

Security Threat 04 is an obvious but critical threat to NGEN operations. Social engineering functions off the goodwill and people-pleasing nature people implicitly have. Due to this, attackers can take advantage of NextGen personnel and their willingness to comply without proper credentials being offered. Security controls to best mitigate this threat include the development of an information assurance training program as well as the introduction of multi-factor authentication when accessing sensitive areas or data.

Sec. Threat 4

Threat ID	Threat(s)	Vulnerabilities	Security Controls
T04	Social Engineering Attack – Exoskeleton Barracks are monitored and patrolled, but personnel are unaware of social engineering techniques. Charismatic threat actors could easily infiltrate sensitive areas.	Lack of personnel awareness and information assurance training – Personnel unaware of human threats when operating on NextGen installations.	1. Implement an information assurance training program (IATP) to educate personnel on cybersecurity awareness. 2. Implement (MFA) to access points throughout NGEN installations containing critical assets.

Security Threat 5

Security Threat 05 has been identified as a serious threat to the business functions of NextGen as an organization due to its prevalence in modern computing and business operations. Malicious software includes ransomware, viruses, worms, and more. This threat takes advantage of the vulnerabilities implicit in human functionality, more specifically our curiosity, by hiding malware within interesting looking emails and messages to gain an entry point into the network by means of an unsuspecting employee. Security controls to effectively mitigate this threat include an

information assurance training program, enterprise level anti-malware, and blacklisting sites deemed not necessary for business functions due to the “awareness” aspect of this threat.

Sec. Threat 5

Threat ID	Threat(s)	Vulnerabilities	Security Controls
T05	Malicious Software Attack – NGEN personnel and employees receive external communications and open them on internal systems. Threat actors could infect NGEN software systems with malicious software.	The opening of unverified emails, email attachments, and enticing links – personnel unaware of proper cybersecurity protocols when dealing with external emails on internal systems.	1. Implement an information assurance training program (IATP) to educate personnel on cybersecurity awareness. 2. Install enterprise level antimalware software on hosts within NGEN networks. 3. Restrict internet usage and site availability on internal systems via firewalls / proxy servers.

Security Threat 6

Security Threat 06 has been identified as a critical vulnerability and threat combination that if realized, could risk the continuity of NextGen. Database data breaches are some of the hardest impacting attacks that could be illustrated by threat actors on a business. They function by means of some initial access into the network and take advantage of weak access controls on the database servers themselves. The security controls of role revocation policies and the encryption of data-at-rest have been identified as the most efficient means to mitigate this threat due to their simplicity and provides a foundation for future threat modeling.

Sec. Threat 6

Threat ID	Threat(s)	Vulnerabilities	Security Controls
T06	Database Breach – NGEN databases are equipped with weak and insufficient access controls. Threat actors could gain unauthorized access to NGEN databases and exfiltrate critical business data.	Weak Database System Access Controls - unauthorized users permitted to view and modify sensitive and classified information.	1. Implement database revocation protocols to revoke access to database information upon transfer or termination. 2. Employ encryption standards for data-at-rest.

Security Threat 7

Security Threat 07 is an extreme threat that requires a high level of threat actor expertise to take advantage of. Session hijacking is an intense attack that requires deep understanding of the session protocols between NGEN systems and Drone Operation Centers. Security controls to mitigate this threat include the adoption of encryption for data-in-transit as well as a token-based authentication system, these have been identified as efficient mitigation techniques where their costs are effective for the asset they protect.

Sec. Threat 7

Threat ID	Threat(s)	Vulnerabilities	Security Controls
T07	Session Hijacking – Exoskeletons operating in the field function remotely via pilot sessions to the DOC. Threat actors could hijack this session and gain remote access and control to exoskeletons.	Lack of encryption of network communications between Drone Operation Centers and NextGen exoskeletons.	1. Employ trusted and tested encryption standards for data-in-transit. 2. Employ a token-based authentication as a means of MFA.

Security Threat 8

Security Threat 08 is a threat that is common in insecure networks that function in ad hoc or mesh modes. The lateral movement of a threat actor through an insecure network relies on weak security controls on mesh nodes and the point of convergence to the main NextGen network. The security controls selected directly mitigate this threat are IDS/IPS systems to detect an attack in progress, and proper network segmentation to separate the networks from each other, limiting the exposure of data to this insecure mesh.

Sec. Threat 8

Threat ID	Threat(s)	Vulnerabilities	Security Controls
T08	Lateral Movement Through Mesh Network – Exoskeletons operating in the field connect to each other and the DOC via an insecure mesh network. Threat actors could gain access to this mesh network and move laterally to gain access to Drone Operations Center and Command Center networks.	Lack of network segmentation between NGEN DOC, Command Center networks, and exoskeleton mesh networks.	1. Properly implement network segmentation to isolate mesh networks and prevent unauthorized lateral network movement. 2. Utilize network traffic monitoring tools, IDS/IPS, and proxy servers to monitor traffic traversing between exoskeleton systems and NGEN operation center systems.

Security Threat 9

Security Threat 09 is not a threat directly undertaken by threat actors, rather it function off the misplacement and loss of a autonomous exoskeleton in drone mode where by attackers can take advantage of the lack of appropriate systems to keep track and monitor the system. Specifically, this threat functions off a lack of geospatial tracking and monitoring systems onboard the NGEN. Security controls to mitigate this threat are very one-dimensional with a tracking system being the

most efficient and effective means of mitigation, while lockdown protocols for drones out of network should be developed to secure local data on NGENs.

Sec. Threat 9

Threat ID	Threat(s)	Vulnerabilities	Security Controls
T09	Loss of Autonomous Exoskeleton – Exoskeletons operating autonomously do not require a physical or remote pilot and as such operate under the control of an artificial intelligence. Threat actors could steal or misplace exoskeletons.	Inadequate geospatial tracking systems onboard NextGen exoskeletons	1. Implement tracking and monitoring systems to track location and status of exoskeletons operating autonomously. 2. Introduce exoskeleton lockdown protocols when connection is lost to DOCs and emplaced in a recovery mode.

Security Threat 10

Security Threat 10 is a common threat to organizations in the modern computing world. Insider threats are employees, personnel, and other actors within an organization who have been wronged, fired, or have been bought by competing companies. These threat actors take advantage of elevated internal privileges and a system with an implicit trust from internal interfaces. Security controls of zero-trust architecture (ZTA) and the adoption of a least-privilege system for user access rights should be adopted as they are the most effective relative to their cost to mitigation effectiveness ratio.

Sec. Threat 10

Threat ID	Threat(s)	Vulnerabilities	Security Controls
T10	Insider Threat – NGEN Employees and personnel are given too many implicit privileges. Disgruntled employees could take advantage of this implicit trust and disrupt	Authorized users given more than needed privileges – implicit trust of the system from users inside the network.	1. Adopt a zero-trust architecture implicitly distrusting internal users and devices.

	business processes.		2. Embrace a least-privilege system to limit assets exposed to internal trusted users.
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Conclusions and Recommendations

Conclusion

The NextGen Exoskeleton Navigator (NGEN) system is a revolutionary technology that will change the way we think about robotics, autonomous systems, and biorobotics. With deployment environments of the best conditions to the worst conditions, the foundations of this system are critical for its success. Presented in this document are ten of the highest risk threats to the NextGen Exoskeleton Navigator system and its deployment group. The safety and success of this project is of the highest priority, and as such has a direct impact on the secure software development lifecycle. The identifying and future modelling of these selected attacks presents an opportunity for our researchers, engineers, and software architects to emplace these security controls as the project is being built to provide the greatest opportunity for success, while mitigating critical vulnerabilities and threats. Security is an everlasting effort in and should be implemented at all stages of development and continued through the lifecycle of the NGEN system regardless of the impact its implementation has on NGENs time to market.

Unaddressed Areas

The selected ten security threats, while broad, do not cover all aspects of the NextGen Exoskeleton Navigator system. Some areas remain vulnerable and should be further evaluated and modelled:

- **Supply Chain Vulnerabilities:** Supply chain attacks could have devastating consequences to the success of the NGEN system, and as such should be reviewed and analyzed swiftly. This document serves as an initial foothold in threat modelling and identification, and due to the nature of external corporations serving as suppliers for NGEN hardware, it is advised that we conduct outreach to them to ensure the security of NGEN hardware procured from outside sources.

- Scientific, Security, and Engineering (SSE) Functions: Functionality relevant to the SSE detachments under the roof of a NGEN deployment should be further evaluated as they are an additional attack surface for threat actors to pivot their way into critical NGEN network systems.
- Maintenance Shop: The maintenance shop for NGENs should be analyzed and modelled as an additional attack surface for threat actors as NextGen Exoskeletons are at their most vulnerable when connected to maintenance crew workstations.
- NGENius: The NGENius logic system should be analyzed and evaluated due to its critical functionality regarding a NGEN system deployments success.

Highest Likelihood Threat

The identification of the highest likelihood threat is difficult to pinpoint due to the volatile nature of NGENs perceived operating environments, however in a generalized perspective the threat with the highest likelihood is security threat (T04) Social Engineering due to its simplicity and effectiveness. This is due to the history of social engineering attacks and their proliferation in modern business operations.

Highest Impact Threat

The highest impact threat possible from these elicited threats is security threat (T06) Database Data Breach, or (T09) Loss of Autonomous Exoskeleton. Two are listed here due to the importance of both assets these threats threaten and the financial repercussions, or the reputational damages received from their realization. The consequences mentioned are best case, with worst case threatening the continuity of NextGen as an organization.

Recommendations:

- **Continuous Vulnerability Analysis:** The concept of security is not something to be done once and forgotten about. Continuous analysis of hardware, software, business processes, system architecture and the threats in which they face should be undertaken.
- **Threat Modeling:** The presented threats within this document are elicited and presented as an initial step towards threat modeling for the NGEN software system. From this, proper threat modeling should be undertaken to identify exact requirements to mitigate threats and patch vulnerabilities.
- **Mitigation Analysis and Implementation:** With threat analysis and identification comes mitigating them. This document was an assessment of threats, leaving the mitigation techniques to be employed in a future report to allow for more NextGen exercises.
- **Incident Response Planning:** Incident response is a critical functionality of business systems. With this, incident response plans should be routinely updated and developed as needed (Adsero Security).
- **Disaster Recovery Planning:** Disasters are always a possibility, and the risk rating of threats will always be non-zero. From this, disaster recovery plans should be developed and consistently revisited to ensure the success and recovery of the NGEN project should disaster strike (Adsero Security).

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